Article

Journal of Sports Economics
2020, Vol. 21(7) 746-780

DOI: 10.1177/1527002520939615
journals.sagepub.com/home/jse
SAGE

Who Should Sign a Professional Baseball Contract? Quantifying the Financial Opportunity Costs of Major League Draftees

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Abstract

Players selected in the annual Major League Baseball First-Year Player Draft have a choice of whether to enter the system or pursue a career outside of professional baseball. As with any choice, there are opportunity costs associated with their decision. The purpose of this study was to analyze those opportunity costs by comparing draftees' expected, non-signing-bonus income to the income of standard laborers. We estimated expected income over the first six seasons of players' careers by combining minor and major league salary data with the probabilities obtained from C5.0 classification models that were applied to a large set of player-related data. Unlike traditional classification methods that produce the probabilities for landing in one of two classes, C5.0 classification models are able to produce the simultaneous probabilities of a player landing in several different classes, or out of professional baseball, in each season. The results show that, all else equal, college fielders drafted in the first 89–90 picks and college pitchers drafted in the first 32 picks (college graduates) or 171 picks (some college) should sign contracts. Pitchers drafted out of high school should sign contracts

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if they are picked in the first 78 picks, as should fielders selected in the top 118 picks. Beyond these ranges, signing bonuses or higher levels of performance are needed to limit the opportunity costs of pursuing professional baseball. Such findings are relevant to draftees making contract decisions and baseball stakeholders dealing with recent calls for improved wages and labor conditions in the minors.

Keywords
opportunity cost, baseball, career progression, C5.0, human capital

Introduction

Each June, approximately 1,200 players are selected in the Major League Baseball (MLB) First-Year Player Draft and presented with the choice to sign with a team, enroll in further schooling, or take a job offer outside of baseball. Those who decide to pursue a career in professional baseball often enter with hopes of making it to the majors and earning a big-league salary that averages between \$4–5 million a season (Berg, 2017). Rookies typically start their careers in Minor League Baseball (MiLB), putting in long hours for little pay under the assumption that better working conditions and larger paychecks are around the corner. However, the reality is the majority of MiLB players do not make it to the majors and will spend their playing career earning an average of just \$7,500 a season in the minors (Berg, 2017). They frequently earn less than minimum wage, have poor medical benefits, need to borrow equipment, and live in shared housing.

In March 2018, Congress made things worse by passing the Save the American Pastime Act, which exempts MiLB players from earning overtime pay if they work more than 40 hours per week. Major League Baseball (MLB) spent approximately \$2 million from 2016-17 in an effort to get the Act passed (Blum, 2018); this occurred despite the fact that MLB player salaries have risen by 10,000% since the start of the free agency era while MiLB wages have lost value against inflation. Rookies earn as little as \$1,160 a month under the new bill, for 3–5 months a year, regardless of how long they are at the clubhouse, field, or gym (Brink, 2018). When considered along with other well-documented hardships of minor league employment (Berg, 2017; Waldon, 2019), it seems as though the majority of MLB draftees should be unwilling to play professionally.

According to Gary Becker's (1975) theory of human capital, athletes' decisions to sign professional baseball contracts are evaluated as investments in human capital. Athletes sign professional baseball contracts because they are willing to forgo income in the present to maximize income from playing Major League Baseball in the future. Scholars have shown, however, that athletes do not always make financially optimal decisions about their careers. From a young age, athletes seem to overestimate their chances of becoming professional athletes (Lee, 1983), and as their sense of themselves as an athlete grows, they are less likely to explore careers outside of sport (Carbrita et al., 2014; Murphy et al., 1996). They may also have coaches, administrators, family, or peers who push them to focus on training and competing instead of developing other options (Horner et al., 2016; Murphy et al., 1996). In short, there are many reasons why athletes might make decisions that do not maximize their future income, like signing a minor league baseball contract with little to no chance of making it to the majors.

In this study, our aim was to analyze the decision to sign a professional baseball contract by predicting the expected income of MLB draftees' over their first six seasons and comparing these figures to the income of non-baseball laborers. Using a large database of player statistics, we used a series of C5.0 classification algorithms to model players' career trajectories over six seasons. C5.0 models are a type of tree or rule-based classification method that can classify observations into more than two categories. As a consequence, C5.0 models produce risk estimates for all possible outcomes that sum to exactly *1*, which is consistent with probability theory and necessary for estimating the risk inherent in baseball careers. In comparison, it is difficult to produce risk estimates that sum to *1* using traditional economic methods such as regression models or basic decision trees. C5.0 models are also more robust to some of the challenges observed by previous scholars studying minor league baseball career advancement; that is, variables predicting success at lower levels do not always predict success at higher levels of the game (Chandler & Stevens, 2012; Longley & Wong, 2011). C5.0 models can be used to overcome these challenges and create unique risk estimates for different players at different levels of the minor league system. By multiplying a player's resulting estimates with minor and major league salary data, we were able to obtain an expected income figure that could be compared to the income of individuals who pursued more traditional career paths.

This article makes three contributions to the literature. First, we use the unique probabilities offered by C5.0 modeling to advance research on minor league player development and athlete career progression. Second, whereas prior research has focused on either pitchers or fielders alone, we analyze the careers of both pitchers and fielders and are able to identify how their decisions to play professional baseball differ. Third, by viewing athlete careers from the perspectives of athletes rather than employers, we apply advanced statistics and financial management best practices to help athletes make better career decisions.

Literature Review

Human Capital Theory

Minor league baseball is a farm system designed to develop players into productive major league players. Therefore, the decisions of athletes and teams can be examined using Becker's (1975) human capital theory, which was designed to explain why people invest in themselves and why businesses train people. Human capital includes knowledge, skills, health, and values embodied in people and not separable from them (Becker, 1975). Organizations and people can invest in human capital and expect long-term returns in income, productivity, political engagement, and quality of life (Psacharopoulos & Patrinos, 2018; Schultz, 1961). Athletes' skills and abilities can also be understood as human capital (Shmanske, 1992). Krautman et al. (2000) were the first to use Becker's human capital theory to analyze minor league baseball as an investment in athletes' human capital.

According to Krautman et al. (2000), a team's minor league system invests in the general training of its athletes as well as firm-specific training. Whereas firm-specific training increases the value of an athlete to his team, general training increases an athlete's productivity (and therefore value) to many teams. Competition from other teams should force the training team to pay a worker equal to his marginal revenue product (MRP), such that athletes' values from general training and their pay rise in tandem. However, if a training team is forced to pay players equal to their MRP, the team would never be able to recoup training costs from them. Becker (1975) initially theorized that firms would share general training costs with workers by paying them less than their MRP during the training period. However, in the case of baseball, Krautman et al. (2000) argued teams recoup their training expenses by extracting surpluses from major league players during the early years of their contracts, when players are not yet eligible for free agency and, as a consequence, are unable to secure competitive salary offers from other teams.

As well as explaining how teams train athletes, human capital theory also explains athletes' decisions to sign professional baseball contracts. According to Becker (1975), people invest in their own human capital to maximize the net value of their future money and psychic income. As a consequence, people are willing to sacrifice income in the present for gains in the future, though at a discounted rate. Apprenticeships and internships are good examples because workers are paid little or nothing in the present, but they expect to receive general on-the-job training that will increase their future income enough to more than compensate for present sacrifices. Since returns to human capital are realized over one's lifetime, younger people are more willing to invest in human capital because they have more years to realize income gains. Athletes similarly view minor league baseball as an investment culminating in the opportunity to sign MLB contracts that will substantially increase their incomes. Players will sign contracts, even if they are aware of poor pay and working conditions in the minors, because they expect to maximize the net value of future income by having an opportunity to play in the majors.

Two variables go into an athletes' net income calculation. First is the future value of income earned by playing in Major League Baseball. By simply making the major league minimum, an athlete will raise his income far above what he was likely to make in another career. The second variable is a risk factor applied to future income streams. The risk factor accounts for the probability of not making it to the majors, which is also large since minor leaguers' likelihoods of making and staying in the majors are relatively slim (Bradbury, 2010; Witnauer et al., 2007). Because few athletes make it, the decision to sign a minor league baseball contract is a human capital investment characterized by high potential income gains but also high risk. Moreover, the chances of making it vary drastically by performance, overall draft pick, position, and whether or not the athlete went to college (Chandler & Stevens, 2012; Spurr & Barber, 1994; Winfree & Molitor, 2007; Witnauer et al., 2007), which means the level of risk will vary by player.

To model the human capital investment decision to sign a minor league baseball contract, a method is needed that can account for the risk inherent in a given players' advancement through the minor league system to the major leagues. The C5.0 method fills this need because it can be used to calculate the simultaneous probabilities of a player (a) making it to the majors, (b) appearing in several different minor league baseball classifications, or (c) falling out of professional baseball, across each season. Moreover, the C5.0 method can produce probabilities that account for differences in performance, overall draft pick, whether or not the athlete went to college, as well as many other variables that relate to career advancement. When combined with salary data, C5.0 probabilities can be used to calculate the net value of baseball players' future income.

Variables Affecting Athlete Career Progression

Prior researchers have examined the variables that influence players' advancement from MiLB to MLB as well as variables that predict performance in the majors. One of the earliest analyses was conducted by Spurr and Barber (1994) as they examined the promotion, demotion, and turnover of minor league pitchers. They found that the time between a pitcher's assignment to one league and promotion or demotion to another (or exit from professional baseball) declined as performance—in terms of *earned runs average (ERA)*, *strikeouts per inning*, and *walks per inning*—deviated from the mean, in either a positive or negative direction. They also used player *age* as a proxy for previous experience, concluding it was negatively associated with the amount of time required to make a determination about a pitcher's ability; however, it did not affect the highest level they were able to reach. Longley and Wong (2011) analyzed over 1,200 pitchers in an attempt to determine the extent to which variations across pitchers in their MLB performances were traceable to corresponding variations in their MiLB statistics. Of the minor league statistics that were examined (*hits*, *walks*, and *strikeouts per nine innings*), only *strikeouts per nine innings* in the minors consistently translated to the majors. Longly and Wong (2011) explained that strikeout production is a physical tool, which is determined earlier in a player's career, and is therefore more likely to translate to later career performance.

Witnauer et al. (2007) looked at career lengths for position players once they reached MLB and found that the average career length was 5.6 seasons, and at every stage of a major leaguer's career his chances of exiting were at least 11%. Bradbury

(2010) further showed that of the hitters who played in the minors between 1992 and 1999, only 28.36% made an appearance in the majors and only 5.10% made at least 1,000 plate appearances. He also suggested that hitters' minor league performances below A+ had little bearing on major league performance, that *strikeout rate* in AAA and AA had the largest effects on hitters' chances of reaching the majors, and that college players selected at the top of the draft generally have higher success rates than high school prospects drafted in similar positions. This echoed the findings of Burger and Walters (2009), who found the expected yields of high school draftees to their teams to be approximately 20% lower than those of college draftees. They also found that drafted pitchers tend to produce lower returns than fielders, and that returns generally decline for late-round selections.

Bierig et al. (2017) used machine learning algorithms to advance on the existing approaches for examining career progression in baseball. Using data from the first six seasons of players' careers, they trained four machine learning models to predict a player's value, as measured by *WAR* (Wins Above Replacement), in years 7–11. The authors were able to predict around 60% of the variation in value for batters but just 30%–40% for pitchers. They concluded that if the decision needs to be made on whether to give a long-term contract to an MLB batter or a pitcher, all else being equal, a team should select the batter. Winfree and Molitor (2007) attempted to estimate the lifetime earnings of position players drafted out of high school with regression models predicting whether a player would receive a signing bonus, how much his bonus would be, and how many years he would spend in the minors and majors. Examining players drafted from 1965 to 1980, their results showed that players drafted in the earlier rounds should enter professional baseball, but players taken in later rounds have higher expected earnings if they attend college. They also found that players who entered professional baseball out of high school spent an average of almost three more seasons in the minors than those who signed out of college. However, their data were dated and only included high school position players. Furthermore, the models only used a player's draft round as a control for his ability and did not differentiate between each minor league classification's different pay scales.

Chandler and Stevens (2012) applied random forest classification models to a dataset of fielders drafted between 1999 and 2002 in an effort to see which MiLB statistics were predictive of MLB success, how these statistics affected a prospect's likelihood of advancement between any two classifications, and how relevant draft position was throughout a player's minor league career. *Strikeouts per at bat* and *strikeouts per walk* in the minors were found to be significantly predictive of success in the majors, and *at bats* was commonly the most important predictor of a player achieving promotion from one classification to another. They further found that the 18th round corresponds to being draft neutral for a team; that is, it represents the point at which draft position no longer matters in determining a player's probability of promotion. While this study and the others we reviewed are helpful for showing which variables may impact player progression between any two classes or to the majors, they are not applicable to a true charting of future progression because they only look at subsets of players who are already present in a given classification and only capture the probability of a player advancing from a lower class to a higher class. In order to examine expected income more accurately in a setting where player movement is not always sequential and statistics have varying influences across different classes, we used the C5.0 models to calculate the simultaneous probabilities of a player appearing in any classification, or falling out of professional baseball, throughout the first six years of his career.

Method

Several steps were used to analyze the decision to sign a professional baseball contract. First, data on professional baseball players were collected. Second, statistical models were created to predict the simultaneous probabilities of a player appearing in different classifications over his early career. Third, financial data were combined with the predicted probabilities in order to estimate players' salary-based incomes through their first six seasons. Lastly, players' incomes were compared to the incomes of people who do not sign professional baseball contracts. Each step is described in the following sections.

Data and Data Collection Procedures

Three different sets of baseball data were collected from The Baseball Cube (*thebaseballcube*.com), a website with extensive historical records of MiLB and MLB team and player statistics that has been referenced in prior research (Chandler & Stevens, 2012). The first dataset consisted of the draft records and biographical information of all players selected from 2003 to 2011 in the MLB First-Year Player Drafts. We selected 2003 and 2011 as boundaries because (a) all players drafted in this timeframe had played out their first contract,[1] (b) they fell within two distinct collective bargaining periods (2003-06 and 2004-11), and (c) they were drafted following the paradigm shift that occurred with the advent of *Moneyball* (Hakes & Sauer, 2006; Lewis, 2003). Data before 2003 may reflect different collective bargaining conditions and market inefficiencies, whereas data after 2011 would have yielded incomplete six-season timeframes. From there, two additional datasets containing seasonal observations of MiLB and MLB player performance statistics (2003-18) were merged with the draft records to form an overall dataset containing the biographical data and first through sixth (if applicable) seasons of playing statistics for players drafted from 2003 to 2011. This dataset was split into unbalanced panels of fielders and pitchers based on players' positions. Players drafted as both pitchers and fielders were split based on the position they ended up playing, and approximately 50 two-way players who switched between pitching and fielding were removed from the original datasets.

Following data collection and splitting, *R* statistical software version 3.5.3 was used to manipulate and clean the data. The following assumptions were used to aggregate the panel data into single, yearly player observations:

- If a player was drafted on more than one occasion, only the information from the draft year in which he actually signed was included.
- If a player appeared for more than one team during a season, his average performance statistics that season were weighted by the number of at bats (fielders) or innings pitched (pitchers) on a team.
- If a player appeared for teams in different classifications within a single season, the highest classification reached was recorded for use as the outcome variable.

By following these procedures, each row contained the highest classification that a player had reached in a prior season, the stats he accumulated that season, and the highest classification he was able to reach the next season. Having the data in this format was necessary to forecast the probability of a player landing in a certain class and achieving the associated payouts over each of his first six seasons. Aside from age, all draft and biographical information remained steady across the players' yearly observations. The full list of variables contained in the final batting ($N =$ 18,646; $n = 3{,}993$; $t = 1, \ldots, 6$) and pitching ($N = 19{,}594$; $n = 4{,}164$; $t = 1, \ldots, 6$) panels, along with their descriptions, can be found in Table 1.

The C5.0 Classification Method

Predicting the career path and projected income of a professional baseball player can be challenging. Each season after being drafted, a player can move between several different professional classifications (Rookie, A-, A, A+, AA, AAA, and MLB) across the minor and major league systems, take a temporary leave of absence (injury, independent league, suspension, etc.), or fall out of the system permanently (retired or cut). Traditional classification methods that sort players into one of two categories cannot capture the full range of possible classifications that exist for professional baseball players early in their careers, and have difficulty accounting for the fact that movement between divisions is not always forward or sequential. Using these methods to obtain the probability of a player making it to a given class or exiting the league would necessitate multiple models for each season, and the summed probabilities of these models would not necessarily equal *1*. Therefore, we employed a classification method that accurately and simultaneously predicted the constrained probabilities of a player reaching all possible categories during a given season.

Classification and regression trees, also known as CART models or decision trees, are popular machine learning methods for analyzing large datasets with a high number of variables and observations. Decision trees can handle variable interactions, multicollinearity issues, and cutoff points with greater ease when compared to traditional regression methods. In this study, an advanced classification method

Table 1. Descriptions of Classifier and Classification Variables.

Variable	Description
Classifier (Predictor) Variables	
Overall	A numerical value indicating the overall draft number at which a player was selected in the MLB First-Year Player Draft; ranges from 1 to at least 1,200. This was the most widely available measure of talent for players drafted out of high school or college.
College	A binary categorical variable indicating whether a player attended college (*1*) for any number of years before being drafted or signed after being drafted straight out of high school (*0*).
Throws	A three-level categorical variable indicating whether a player throws right handed, left handed, or both.
Bats	A three-level categorical variable indicating whether a player bats right handed, left handed, or as a switch-hitter (both).
Height	A player's height, in inches, at the time of being drafted.
Weight	A player's weight, in pounds, at the time of being drafted.
PriorClass	A six-level categorical variable showing the highest classification a player was able to achieve in a prior season; recorded as *HalfSeason* for the Rookie and A- classifications, *FullA* for A and A+, *AA*, *AAA*, and *MLB*. If a player did not play in a season *t*-2 or earlier but later returned to professional baseball, he was categorized as *DNP* for that season.
PerfStats	A vector of performance statistics related to fielders or pitchers. For fielders, the following prior-season stats were included: on-base percentage (*OBP*), slugging average (*SLG*), at bats (*AB*), and strikeouts per at bat (*SO/AB*). For pitchers: walks and hits per inning pitched (*WHIP*), earned runs average (*ERA*), strikeouts per nine innings (*SO9*), innings pitched (*IP*), and the percentage of appearances in which the pitcher started the game (*Start*%) or closed the game (*Close*%). These statistics were chosen for their relevance in accurately quantifying the batting and pitching performances of the players, as highlighted in prior literature (Beneventano et al., 2012; Chandler & Stephens, 2012; Longley & Wong, 2011; Spurr & Barber, 1994). If a player appeared on multiple teams during the same season, the weighted averages of these stats were calculated using at bats or innings pitched.
Age	A player's age, in years, during the prior season.
Classification (Outcome) Variable	
Class	A factor vector with six levels representing the highest professional baseball classification (*HalfSeason, FullA, AA, AAA, MLB*) that a player was able to reach in an observed season, or that it was the first season in which the player was permanently out of professional baseball (*OUT*).

known as C5.0 was implemented using *R* statistical software version 3.5.3. This method has been successfully implemented in other business settings, such as supply chain management (Emerson et al., 2009) and classifying consumers into different market segments (Chiang, 2012). As a machine learning method, C5.0 extracts informative patterns from data and attempts to predict a case's class from a set of associated classifier variables (Kuhn & Johnson, 2013; RuleQuest, 2019). Unlike traditional CART models, random forests, and most other decision tree methods, C5.0 is able to classify an observation into more than two classes. It does this using either tree-based or rule-based algorithms that split the sample based on variable attributes that reduce uncertainty and provide higher ratios of information gain. Information gain ratios essentially display how much information is added from a split relative to the information already present in the data (Quinlan, 1986). By splitting on attributes that provide more information about (shared dependence with) an outcome class, attribute homogeneity within all classes can be increased. Beginning with an entire set of data, the sample is first split based on the variable providing the most information gain. For example, draft position was typically the most informative variable for classifying a player during their first season, so the C5.0 model would first split the season-one observations on one attribute of the draft variable (e.g., drafted in the top 300 picks). Further splits based on combinations of other variables and attributes follow until terminal predictions for each observation are achieved. C5.0 is also open to boosting, a procedure through which multiple models are built in sequence, each with the goal of improving on the errors of the prior model. The boosted model is developed via a voting procedure that combines the models' separate predictions into one final prediction that should have a reduced error (misclassification) rate. The model iterations used in the boosting procedure are referred to as trials.

Empirical Specifications

Across both positional categories, the C5.0 models consisted of six ($t = 1, \ldots, 6$) variations of the following base model:

$$\begin{aligned}\Pr(\mathrm{Class_{it}}) = {} & \mathrm{Overall_i} + \mathrm{College_i} + \mathrm{Throws_i} + \mathrm{Bats_i} + \mathrm{Height_i} + \mathrm{Weight_i} \\ & + \mathrm{PriorClass_{it-1}} + \mathrm{PerfStats_{it-1}} + \mathrm{PriorClass_{it-2}} + \mathrm{PriorClass_{it-3}} \\ & + \mathrm{PriorClass_{it-4}} + \mathrm{PriorClass_{it-5}} + \mathrm{Age_{it-1}},\end{aligned}$$

where the probability of player *i* reaching a given *Class* during season *t* is predicted by his overall draft number, whether he played in college or not, what hand he threw and batted with, his height, his weight, the highest class he had reached during the prior season, his relevant performance stats from that prior season, the highest classes he had achieved in any other prior seasons (applicable to seasons 3 through 6), and his age during the prior season. The season-one classification models differ from the other seasonal models in that the prior performance and prior classification variables were not available for inclusion.[2]

The *Class* variable is a multi-level outcome variable consisting of the five (season one) to six (seasons two through six) different categories in which a player observation could be classified each year. However, unlike traditional models, C5.0 ensures that a player's probabilities of landing in these different categories sum to *1*. The categories present in this variable were formed through a combination of the seven different professional baseball classes (Rookie, A-, A, A+, AA, AAA, and MLB) and an *OUT* category marking the first season in which a player was completely out of the system. Because we only analyzed draftees who started a professional baseball career, the possibility of a player being classified as *OUT* (i.e., never playing at any level after being drafted) is not allowed in the year-one models.

In an effort to improve classification simplicity and accuracy, we aggregated the Rookie and A- leagues into a *HalfSeason* classification because players in these divisions earn similar pay and play shorter seasons. The A and A+ classes were also grouped together beneath the *FullA* label because they operate on near-identical pay scales and are both full-season, single-A leagues. All other classes (*AA*, *AAA*, *MLB*, and *OUT*) remained separate due to their more notable pay and performance differences. The sample size available for each subsequent model shrunk after removing players who had been classified as *OUT* in a prior season. Players who were not finished with their careers, but who did not play (*DNP*) in MiLB or MLB during an observed season or the prior season (e.g., they were injured, suspended, taking time off, or playing in a foreign or independent league), were excluded from the samples that were analyzed in those seasons. If they returned to the system and their *DNP* designation occurred in seasons *t-2* to *t-5,* they were included with *DNP* as a level of the *PriorClass* variable.

In total, there were six different models for pitchers and fielders, each representing the first six years of a player's career. Though the data could extend beyond this threshold, the sixth season was set as the limit because minor leaguers not on a 40-man MLB roster by that stage become minor league free agents. Furthermore, the majority of players are either out of the league or playing in the majors by their sixth seasons. This can be seen in Table 2, which shows the percentage of fielders and pitchers drafted and signed from 2003 to 2011 present in each classification from the first through sixth seasons of their careers. Table 2 shows that, across drafted pitchers and fielders, most baseball players are either out of the system or in the majors by year six. More specifically, nearly 39% of fielders are out of the league by year six while 26% are in MLB. The majority of pitchers are out of the league by year six (64%), with 13% in MLB. Therefore, six seasons was an appropriate cutoff for examining the draftees' expected incomes.

Model Tuning, Cross Validation, and Variable Importance

A benefit of the C5.0 method is that it can be tuned using a series of cross validation procedures to recommend an ideal number of boosting trials and decide whether the final model should be winnowed to remove unhelpful variables and avoid

Table 2. Classification Progressions of Drafted Fielders and Pitchers over their First Six Seasons.

Classification	Season 1	Season 2	Season 3	Season 4	Season 5	Season 6
Fielders (n = 3,993)						
Rookie and A-	81.5%	22.4%	7.0%	1.2%	0.6%	0.1%
A and A+	16.0%	62.8%	47.1%	26.3%	10.0%	4.6%
AA	1.5%	7.5%	23.2%	25.4%	19.9%	11.9%
AAA	0.8%	2.3%	5.5%	11.9%	16.1%	17.4%
MLB	0.2%	1.0%	4.8%	13.3%	21.3%	25.8%
DNP	0.0%	1.3%	1.3%	1.4%	1.7%	1.5%
OUT	0.0%	2.7%	11.1%	20.5%	30.4%	38.7%
Pitchers (n = 4,164)						
Rookie and A-	81.3%	22.3%	7.4%	2.1%	0.9%	0.6%
A and A+	16.4%	52.7%	35.6%	16.8%	7.4%	3.1%
AA	1.2%	6.9%	15.8%	17.1%	12.1%	7.2%
AAA	0.7%	2.4%	5.2%	8.3%	9.6%	10.0%
MLB	0.4%	1.4%	3.7%	7.6%	12.2%	13.2%
DNP	0.0%	3.2%	3.0%	3.0%	2.2%	1.9%
OUT	0.0%	11.1%	29.3%	45.1%	55.6%	64.0%

overfitting. It can also generate rulesets that prune the model using independent conditional statements that are easier to interpret than decision trees, especially in models with a large number of observations and variables (Kuhn & Johnson, 2013; RuleQuest, 2019). Each model was tuned to determine which elements allowed it to achieve the highest degree of accuracy through cross validation. This procedure subjected the models to repeated *k*-fold cross validation whereby the observations were shuffled and split into 10 groups (i.e., folds). Each fold was alternatingly withheld as testing data while the models were trained on the remaining nine groups of data. This process was repeated five times for each model, and the model type with the highest accuracy on the testing folds was retained and used. The tuning parameters, in-sample error (misclassification) rates, and cross validation (out-of-sample) accuracies of the selected models are included in Table 3.

Even though they were tasked with classifying a player in one of six possible categories, the models correctly classified the majority of the out-of-sample test data. The κ values, which measure observed accuracy compared to expected accuracy, were almost all within the fair to moderate range (Landis & Koch, 1977). These findings confirmed the method was adequately reliable for predicting the probabilities of a player reaching a certain class in his first six seasons. Nonetheless, though C5.0 was a relatively accurate and more appropriate method given the wide array of potential classifications, its method of repeatedly splitting and subsetting the data makes precise interpretations of variable effects more difficult.[3] Fortunately, we are still able to see which classifier variables play a consistent role in determining an observation's class through the percentage of splits associated with each variable.

Table 3. C5.0 Model Parameters for Fielders and Pitchers.

Model (*n*)		Type	Trials	Winnow	Error[a]	Accuracy (κ)[b]	
Fielders							
Y1	*(3,993)*	Tree	20	False	15%	85%	*(.038)*
Y2	*(3,930)*	Rules	20	False	9%	62%	*(.367)*
Y3	*(3,375)*	Rules	20	False	2%	55%	*(.371)*
Y4	*(2,610)*	Rules	20	False	2%	52%	*(.380)*
Y5	*(2,026)*	Rules	20	False	1%	53%	*(.413)*
Y6	*(1,561)*	Tree	20	False	0%	57%	*(.435)*
Pitchers							
Y1	*(4,164)*	Tree	20	False	17%	82%	*(.147)*
Y2	*(4,031)*	Rules	20	False	6%	58%	*(.312)*
Y3	*(3,462)*	Rules	20	False	1%	52%	*(.346)*
Y4	*(2,726)*	Rules	20	False	1%	50%	*(.366)*
Y5	*(2,104)*	Rules	20	False	1%	51%	*(.378)*
Y6	*(1,696)*	Rules	20	False	2%	55%	*(.413)*

[a] in-sample misclassification rate; [b] out-of-sample, 50-fold cross validation results.

Table 4. Variable Importance Scores: Fielder Models.

	Y1	Y2	Y3	Y4	Y5	Y6
Age	31.25%	15.29%	10.89%	7.98%	9.82%	7.03%
Bats	1.25%	5.23%	5.36%	4.68%	4.41%	4.56%
Throws	7.5%	2.96%	2.35%	2.53%	2.60%	4.06%
Overall	40%	17.08%	12.05%	8.82%	5.24%	7.47%
Height	10%	7.96%	5.77%	5.69%	5.27%	7.85%
Weight	10%	6.28%	6.01%	5.22%	4.32%	8.30%
College	0%	1.55%	2.89%	3.24%	3.48%	2.44%
OBP	NA	7.65%	7.14%	10.62%	7.80%	7.97%
SLG	NA	7.69%	8.91%	7.07%	8.83%	8.15%
AB	NA	12.16%	15.35%	12.77%	13.51%	8.70%
SO/AB	NA	8.18%	6.92%	6.07%	4.12%	7.85%
Y1 Class	NA	7.99%	4.97%	4.80%	4.13%	4.21%
Y2 Class	NA	NA	11.41%	7.04%	6.00%	4.91%
Y3 Class	NA	NA	NA	13.47%	7.36%	6.03%
Y4 Class	NA	NA	NA	NA	13.11%	6.13%
Y5 Class	NA	NA	NA	NA	NA	4.36%

Note: Importance scores represent the percentage of splits associated with a variable.

These percentages, known as importance scores, are reported for the fielder and pitcher models in Tables 4 and 5, respectively.

The importance scores show that certain predictors, such as draft position and prior classes predating the most recent one, logically diminish in importance as the

Table 5. Variable Importance Scores: Pitcher Models.

	Y1	Y2	Y3	Y4	Y5	Y6
Age	34.18%	13.04%	8.35%	9.18%	7.61%	11.20%
Bats	8.23%	4.44%	3.83%	3.11%	2.95%	2.19%
Throws	2.53%	2.72%	2.20%	1.74%	1.93%	1.96%
Overall	34.18%	15.93%	8.90%	5.56%	5.46%	2.97%
Height	10.13%	6.75%	6.22%	6.29%	3.81%	4.50%
Weight	10.13%	5.38%	4.83%	4.46%	3.65%	3.49%
College	0.63%	1.52%	3.70%	1.92%	2.76%	1.66%
WHIP	NA	10.62%	8.17%	8.02%	8.61%	8.34%
ERA	NA	6.97%	7.39%	7.49%	7.41%	7.61%
IP	NA	7.47%	12.46%	9.50%	9.70%	9.76%
SO9	NA	7.30%	6.26%	5.89%	6.63%	4.94%
Start%	NA	4.28%	6.32%	7.00%	5.30%	3.39%
Close%	NA	3.58%	3.89%	3.59%	3.44%	3.05%
Y1 Class	NA	10.01%	5.19%	5.00%	4.51%	3.06%
Y2 Class	NA	NA	12.30%	7.63%	6.13%	5.23%
Y3 Class	NA	NA	NA	13.62%	7.06%	6.04%
Y4 Class	NA	NA	NA	NA	13.08%	5.49%
Y5 Class	NA	NA	NA	NA	NA	15.12%

Note: Importance scores represent the percentage of splits associated with a variable.

seasons progress. Others, such as player handedness, height, and weight, also take on less importance compared to other variables as a draftee's career progresses. As these effects begin to diminish, key performance stats such as slugging average (*SLG*) and on-base percentage (*OBP*) in fielders, and *WHIP*, *ERA*, and *SO9* in pitchers, begin to take priority in determining where players end up. Fittingly, the variables most frequently used for categorizing the players were at bats (*AB*) or innings pitched (*IP*), and the highest division a player had reached in the prior season. Opportunity at a given level, it seems, was fairly consistent in classifying a player's arrival to a future class.

Model Application Procedures and Salary Data

The models calculated the constrained probabilities of a professional baseball player reaching any number of six different classes (*HalfSeason, FullA, AA, AAA, MLB,* or *OUT*) in the first six seasons of his career. Multiplying available salary data by these probabilities, the models projected how much a player would earn over his first six seasons, exclusive of signing bonuses. Salary data were obtained from a sample of 2017 MiLB player contracts that were shared with us by former MiLB players. These contracts showed that the standard monthly pay for players was approximately \$1,250 in half-season leagues (Rookie and A-), \$1,500 in A and A+, \$1,800 in AA, and \$2,250 in AAA. All full-season classifications operate on five-month pay schedules, while the half-season leagues pay their players over three months. The 2017 league minimum salary of $535,000 was assumed for players classified in MLB. All of the baseball-related financial figures were taken from 2017 in order to match the 2017 census data that was used to calculate the income of non-baseball individuals.

The average yearly incomes of 18–24 year old males who did not pursue professional baseball were collected from the U.S. Census Bureau's (2017) statistics and compared to the predicted salaries of the players in order to calculate the financial opportunity costs of pursuing baseball over a traditional career path. The total, six-year incomes were $147,886 for 18–24 year olds with a high school diploma, $116,539 for 18–24 year old males who had attended some college or completed a two-year associate's degree,[4] and $231,973 for 18–24 year old males with a four-year bachelor's degree. The three totals began with annual income of $23,268, $18,336, and $36,498, respectively, and assumed the national average growth rate of 2.3% that occurred in income from 2006 to 2017 (Bureau of Labor Statistics, 2019).

To outline how predicted incomes were calculated for players, consider the following example: Player A was a college pitcher drafted No. 1 overall with average statistics across all other variables. According to the year-one C5.0 model, he had a 29% probability of half-season ball being the highest classification reached, a 61% probability of landing in A or A+, a 6% probability of reaching AA, and a 4% probability of reaching MLB in his first season. These probabilities were then multiplied by the respective salaries for each classification and summed to provide an estimate of what the player's income would be in year one ($25,705, excluding any signing bonus). This process was repeated using the models for each subsequent season, with the sum total representing how much he was expected to earn from seasons one through six of his career. Players who were classified as *OUT* in a given year (i.e., the highest probability was *OUT*) took on the average salary of non-baseball individuals, with the ascending 2.3% raise implemented over their remaining *OUT* seasons. For the *PriorClass* variables, the classification with the highest probability of occurrence was employed.

Results

The Effect of Overall Draft Position on Predicted Income

In order to assess the impact that draft position has on projected income, we simulated a typical draft order ($Overall = 1, \ldots, 1{,}200$) while holding the values for all other variables constant at their seasonal means. In this first analysis, we only examined players who had attended or graduated from college. Table 6 displays the mean values for the variables that were held constant. The predicted, six-year payouts for average fielders selected with the 1–1,200 overall picks fell within 138

Table 6. Mean Values for Quantitative Player Variables by Season.

Fielders	Height	Weight	Age	OBP	SLG	AB	SO/AB		
Y1	73	194	21	NA	NA	NA	NA		
Y2	73	194	21	0.333	0.359	183	0.238		
Y3	73	194	22	0.329	0.370	324	0.238		
Y4	73	194	22	0.325	0.373	373	0.236		
Y5	73	194	23	0.325	0.379	394	0.232		
Y6	73	194	24	0.326	0.386	393	0.230		
Pitchers	**Height**	**Weight**	**Age**	**ERA**	**WHIP**	**Start%**	**Close%**	**IP**	**SO9**
Y1	75	199	21	NA	NA	NA	NA	NA	NA
Y2	75	199	21	4.215	1.415	0.35	0.23	45	8.543
Y3	75	198	22	4.303	1.423	0.44	0.23	93	7.873
Y4	75	197	23	4.393	1.439	0.45	0.21	103	7.654
Y5	75	197	24	4.388	1.437	0.43	0.22	107	7.628
Y6	75	197	24	4.357	1.437	0.40	0.22	105	7.636

Notes: Most players attended college and were right handed; the class that had the highest probability of being reached in a prior season was used as *PriorClass* in the next model.

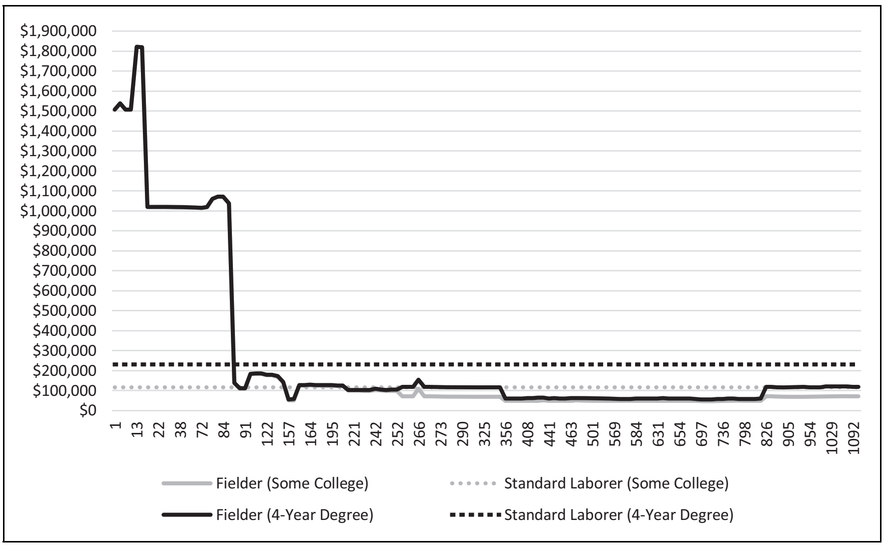

Figure 1. Projected 6-year earnings of former college fielders by overall draft pick.

distinct tiers, ranging from a maximum of $1,821,628 for those drafted in slots 13–15, to $55,585 at picks 705–714. For average pitchers, the projected payouts fell within 126 distinct tiers, ranging from $1,062,105 at picks 4–10, to $63,311 at slot 193. It is important to remember that these projections do not include signing bonuses. Figure 1 and Figure 2 display these results for college fielders and pitchers,

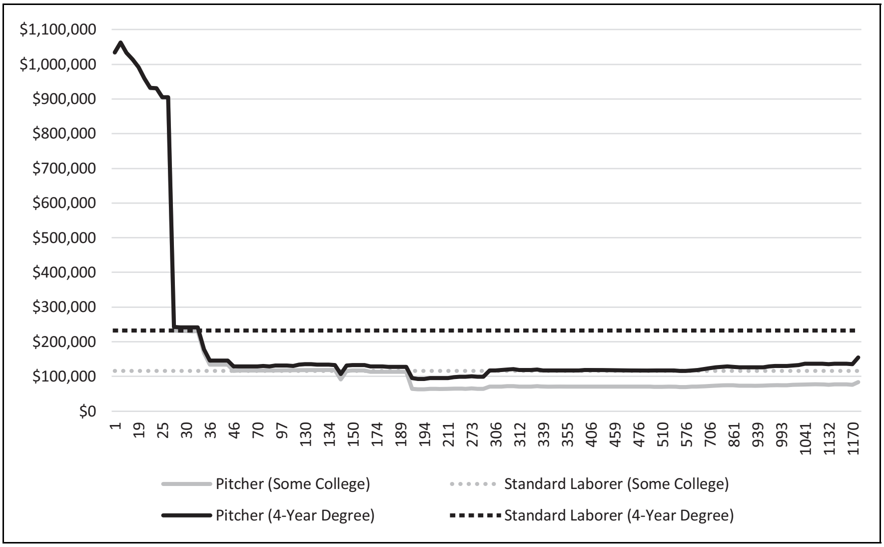

Figure 2. Projected 6-year earnings of former college pitchers by overall draft pick.

respectively, along with dashed, horizontal lines indicating the projected six-year incomes of 18–24 year old males with four-year degrees or some college.

Fielders drafted before the 89th overall pick (90th overall pick for those with some college) increase their projected six-year income compared to what they would make in the workforce. However, after these slots, the decision becomes more complicated. For fielders who graduate college, annual baseball income falls below the average income level at pick 89, indicating that fielders taken after the third round will need a signing bonus or other source of income to compensate for the income they are sacrificing by pursuing baseball. Those who were drafted with an associate's degree or some college credit fall beneath the line at pick 90; however, this difference is marginal ($6,165) and subsequent picks return above this level at various points. This trend of hovering around the breakeven point continues until pick 205 (rounds 6–7), where the income of the player drops and remains below the six-year income of a non-baseball individual with some college credit.

It should be noted that an average-performing fielder is projected to be out of professional baseball by his fourth season if he was taken between picks 253 to 355 or 826 to 1,200. Interestingly, an average fielder taken between picks 356 and 825 (rounds 12–27) was predicted to remain in professional baseball, but his income ended up being less than those who dropped out. This is due to players who drop out earning the annual non-baseball wage for the remainder of the six-year cycle, while holdouts are stuck with low-classification, minor league wages. Even in years five and six, these holdouts' highest probabilities were consistently in A to A+ (*FullA*), and they were never projected to make it to the majors.

Switching to pitchers, Figure 2 shows an even sharper drop-off in the income of these players as they fall in the draft. This coincides with the survival rates that were seen in Table 2, which showed that pitchers face a steeper challenge to make MLB rosters. The common pitcher with a college degree will drop below the expected income of an equally educated non-baseball laborer at pick 33 and never recover. For the average college pitcher drafted before obtaining a four-year degree, picks 140–148 are expected to dip just below the non-baseball income level before picks 149–171 rise slightly above the breakeven point. Pitchers drafted 172nd overall or later are expected to fall and remain below the income threshold of similarly educated, non-baseball laborers in their first six years of employment.

Unlike the fielder projections that had a segment of late-round draftees who were projected to remain in the minors longer than they should have, the pitching projections show a consistent trend of players dropping out of professional baseball the later in the draft they are selected. An average pitcher selected from picks 1,177 to 1,200, for example, was expected to be out of the system by his third season. Picks 304 to 1,176 were projected to be out of professional baseball by their fourth seasons, and picks 192 to 303 were predicted to be finished by year five. Similar to fielders, late round picks who dropped out a season earlier were expected to accumulate higher six-year income totals than those who waited another season to try and make it. Again, this is due to their income as standard laborers being higher than their projected, non-bonus income in professional baseball.

The Effect of Signing out of High School on Predicted Income

To assess the opportunity cost of signing from high school, the draft position models were reemployed to include the *College* variable with a value of *0* and *Age* variables that were reflective of the average age of players who did not attend college in each year. The average ages ranged from 18 in the first season to 23 in the sixth. All other variables were held constant at the previously stated seasonal means. Figures 3 and 4 show the results for fielders and pitchers, respectively.

Figure 3 shows that, all else equal, a fielder who signs straight out of high school is expected to generate an income greater than that of the standard laborer through pick 118 (the end of round four). Barring a brief return at picks 139 and 140, picks 119 and beyond are expected to net the draftees less than they would make had they sought employment outside of professional baseball. On average, the 1,080 draftees below the threshold are expected to sacrifice a total of $39,468 over their first six seasons. The steep drop at the start of the line in Figure 3 is evidence of the top two picks being the only ones projected to make it all the way to the majors in their first six years. Though the other picks above the threshold have sizable probabilities of reaching MLB during this term, the remaining draftees have little to no chance of plying their trade at the game’s top level.

However, it should be noted that none of these early entrants were projected to be *OUT* by the models; that is, if a young, late-round fielder performed at the average

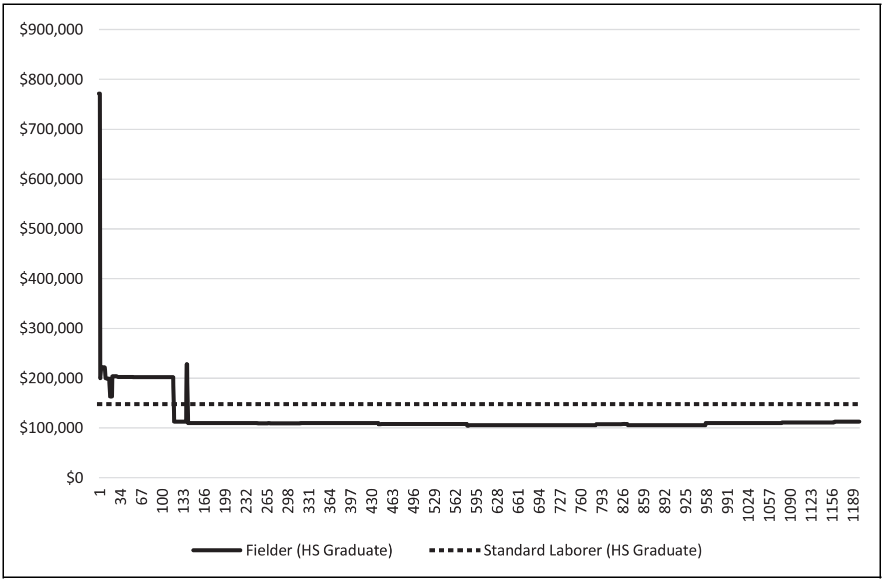

Figure 3. Projected 6-year earnings of high school graduate fielders by draft pick.

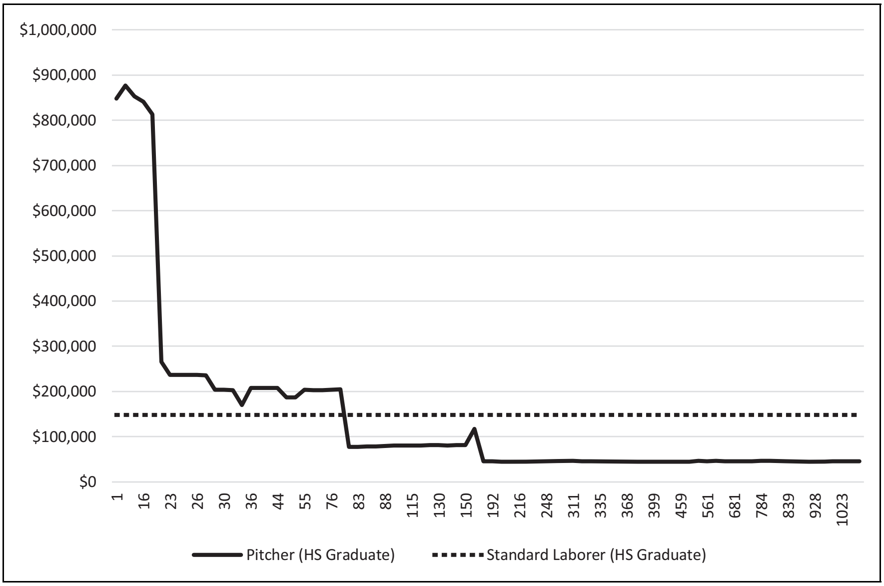

Figure 4. Projected 6-year earnings of high school graduate pitchers by draft pick.

level at the plate in each of his first six seasons, he was able to maintain his dream of playing professional baseball. While this may seem unreasonable, Winfree and Molitor (2007) noted that fielders drafted out of high school remained in the minors for 4.38 years, or approximately 3-years longer than those drafted out of college. They attribute this to a selection problem whereby players who enter the draft straight out of high school have more inherent talent—and therefore more staying power—than players who attended college because they were not good enough to get drafted earlier. As a result, teams may be more willing to give these younger prospects additional seasons to develop their physical and technical hitting tools. We certainly find evidence of this, as high school draftees being held at the same level of performance as college draftees were able to remain in professional baseball over a longer period of time and out-earn their non-baseball counterparts over an extended range of draft picks.

Shifting to the pitchers, Figure 4 visualizes their six-year incomes compared to the income generally accrued by high school graduates who did not pursue baseball. Compared to the average high school fielder, there was a shorter range of draft selections before the income of the average pitcher was expected to dip below the income of the standard laborer. Indeed, it was during the middle of the third round (No. 79 overall) that the income of a pitcher drafted straight out of high school was expected to drop below the projected six-year income of the non-baseball worker ($77,851 compared to $147,886). From this point onward, the average pitcher is projected to accumulate approximately $98,000 less than he could earn working a normal job outside professional baseball.

The only high school pitchers predicted to make it all the way to the majors in their first six seasons while performing at the mean are those taken with a top-19 pick. However, like the fielding models, the pitching models did not predict high school pitchers to be fully *OUT* during this timeframe. At both positions, teams appear more willing to develop younger, high school prospects than college draftees. Nonetheless, the majority of draftees would be better off, financially, if they took up different careers. Though not predicted to be out of the system, their chances of making it to the MLB are slim for all but the top few.

The Effect of Performance Changes on Predicted Income

So far, player performance statistics have been held constant. In this section, key performance variables were manipulated to examine their effects on the early-career incomes of players who graduated from college. As was shown in Table 4 and Table 5, C5.0 models produce variable importance scores based on the percentage of splits associated with a given predictor variable. Across the five seasons with performance statistics, on-base percentage (*OBP*) was consistently one of the more important performance statistics for classifying fielders, while strikeouts per nine innings (*SO9*) has been recognized in prior literature as a major league indicator and a variable over which pitchers have more direct control. As such, the next set of

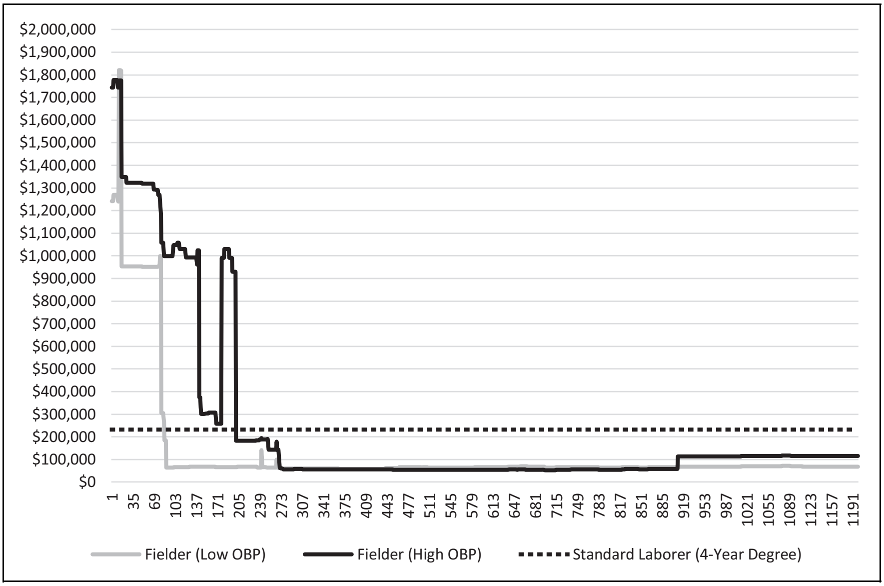

Figure 5. Projected 6-year earnings of fielders (4-year degree) by draft pick and OBP.

analyses demonstrated what would happen if a player were to appear in low performance or high performance categories—as judged by the quartiles present in the data each season—while also altering draft position and holding all other variables constant at the means presented in Table 6.

Looking first at the fielders, players with lowest quartile OBP values (25th percentile) were predicted to accumulate average, non-bonus incomes of $118,353 or $131,908, depending on whether they had some college or a college degree, respectively. For players with highest quartile OBPs (75th percentile), these values were $224,756 and $240,451, respectively, equating to average differences of $106,402 (some college) and $108,543 (college graduates) between them and identical players with lower OBPs each season. The annual OBP values ranged from .297 to .301 for the low performers and from .358 to .372 for the high performers. Figure 5 summarizes these findings, showing that college graduate fielders who consistently get on base at a rate that ranks in the top quartile are able to avoid dropping below the standard income threshold until the 201st pick. Concurrently, those who consistently rank at the bottom quartile for OBP drop below that level at pick 86.

As for the pitchers, the higher performing (75th percentile) SO9 values ranged from 8.802 in year four to 10.002 in year one, while the lower performing (25th percentile) SO9 values ranged from 6.182 in year five to 6.9 in year one. These ranges align with the prior literature suggesting that strikeouts, as a physical tool, manifest early in a pitcher's career (Longley & Wong, 2011). The high performers

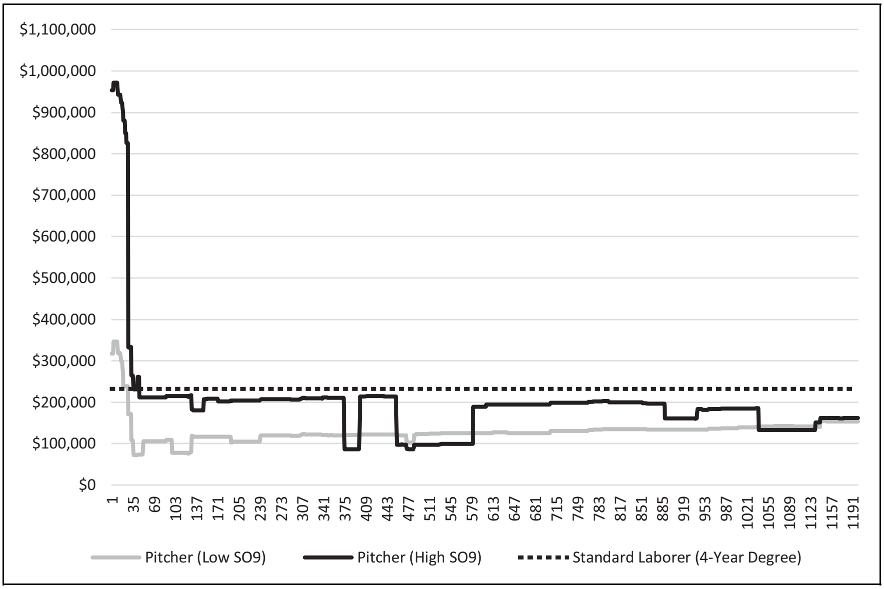

Figure 6. Projected 6-year earnings of pitchers (4-year degree) by draft pick and SO9.

who had attended some college were predicted to make an average of at least $92,960 more ($15,493 per season) than those who were throwing fewer strikeouts per nine innings. Pitchers who were college graduates stood to make an average of $66,204 more ($11,034 per season) for being in the top-performing quartile as compared to the bottom quartile. The differences for pitchers with college degrees are seen in Figure 6, where the projected six-year incomes are once again contrasted to the average six-year income of non-baseball individuals.

As seen, the income potentials vary for the high and low performers, suggesting that throwing more strikeouts can increase a pitcher's likelihood of accumulating more income than a standard laborer. Players with better SO9 s are generally able to out-earn the non-baseball average for college graduates through the 46th pick in the draft, whereas those with worse SO9s fall below the threshold after pick 28 and never recover. Some pitchers with poor SO9s were also predicted to be out of the league as early as their third year; for those with good SO9s, being out was delayed until the fourth or fifth season, and only for those selected after pick 1,040.

Analysis of Player Ideal-Types

As one final mode of analysis, we examined how four different player types from each position were projected to progress through their first six seasons. This analysis exemplifies the main strength of C5.0 analysis—analyzing individual players' prospects and producing individualized risk assessments. These simulations collated the results from the first three analyses with one addition: we assume a player will perform at the average of players *like him,* rather than assuming he will perform like the average of all players (or the top or bottom quartiles of players) in a given season. This accounts for the likelihood that players drafted at different picks will perform differently. First, we inputted a hypothetical college graduate drafted No. 10 overall (21 years old at draft time), with average height, weight, and handedness, into the C5.0 model for season one. For season two, we set the player's performance at the average of first round picks' playing statistics. Every season thereafter, we used the average performance statistics from the class the player was projected to have played in the season before. We repeated this process for hypothetical high school draftees (18 years old at the time of the draft) also selected No. 10 overall, and for college and high school graduates drafted No. 300 overall. Age increased sequentially with each yearly model from 21 to 26 for college draftees and from 18 to 23 for high school draftees. Table 7 displays the probabilities obtained for each class in each season, along with cumulative calculations of their non-signing bonus income and cells showing the differences between their expected incomes and those of the standard laborer. As before, when a player was classified as out, he assumed the standard annual income of a similarly educated, 18–24 year old male.

Looking first at a hypothetical college graduate pitcher drafted No. 10 overall, we see that his draft position and biographical characteristics classify him with the greatest probability (62%) of playing in A or A+ in season one. If he then performs at a level consistent with other first round pitchers, he increases his chances of reaching MLB in year two by almost 47%. Having reached the majors by his second season, he can maintain his stay in the highest division, and increase his income, by continuing to perform at a league-average level or better over the remaining four seasons. Shifting to an identical pitcher selected straight after high school, we notice a few differences over his first few seasons. First, this pitcher is much more likely to land in the Rookie or A- classes during his initial season, which is to be expected given his age and development. However, if he performs at a level consistent with first round draft picks during his inaugural season, he will advance to A or A+ the following season. Then, average single-A performance will see him advance to AA in year three before making the jump to the big leagues in year four.

Pitchers who were drafted No. 300 overall have far different prospects, despite having the same biographical statistics. The 300th-pick college graduate was projected to be out of professional baseball by year four, having forgone approximately $104,000 after performing at levels consistent with 10th and 11th round draft picks in year one, and single-A pitchers during seasons two and three. The high school pitcher was able to stay around a little longer, but with just average levels of performance across the half season and single-A leagues, he was trending toward being out with no chance of making the majors. Most worrisome in this scenario is the roughly $97,000 he sacrifices by holding out in the minors.

Shifting to fielders, we see that a college graduate selected with the 10th pick in the draft is expected to make the majors by year two or three and remain there if he

Table 7. Classification Probabilities and Cumulative Income ($) Projections Using Player Ideal-Types.

PITCHERS	Y1	Y2	Y3	Y4	Y5	Y6	Diff.	FIELDERS	Y1	Y2	Y3	Y4	Y5	Y6	Diff.
Rookie/A-[10]	29%	0%	0%	0%	0%	0%		Rookie/A-[10]	43%	0%	0%	0%	0%	0%	
A/A+	62%	19%	0%	0%	0%	0%		A/A+	57%	10%	0%	0%	0%	0%	
AA	6%	24%	0%	0%	0%	0%		AA	0%	48%	19%	0%	0%	0%	
AAA	0%	6%	0%	0%	0%	5%		AAA	0%	0%	15%	0%	3%	0%	
MLB	3%	50%	100%	100%	100%	95%		MLB	0%	42%	66%	100%	97%	100%	
OUT	NA	0%	0%	0%	0%	0%		OUT	NA	0%	0%	0%	0%	0%	
Cum. $	*22.3k*	*294k*	*829k*	*1.36m*	*1.9m*	*2.41m*	*+2.2m*	*Cum. $*	*$5.89k*	*236k*	*592k*	*1.13m*	*1.65m*	*2.18m*	*+1.9m*
Rookie/A-[10]	86%	18%	0%	0%	0%	0%		Rookie/A-[10]	95%	0%	4%	0%	0%	0%	
A/A+	8%	68%	34%	0%	0%	0%		A/A+	5%	100%	56%	29%	0%	0%	
AA	4%	14%	60%	36%	0%	0%		AA	0%	0%	28%	24%	24%	0%	
AAA	0%	0%	0%	14%	2%	5%		AAA	0%	0%	0%	47%	32%	9%	
MLB	2%	0%	6%	50%	98%	95%		MLB	0%	0%	12%	0%	44%	91%	
OUT	NA	0%	0%	0%	0%	0%		OUT	NA	0%	0%	0%	0%	0%	
Cum. $	*14.9k*	*21.9k*	*62k*	*334k*	*859k*	*1.37m*	*+1.2m*	*Cum. $*	*3.93k*	*11.4k*	*82.5k*	*92.1k*	*333k*	*821k*	*+673k*
Rookie/A-[300]	88%	0%	0%	0%	0%	0%		Rookie/A-[300]	84%	0%	0%	0%	0%	0%	
A/A+	12%	93%	56%	9%	0%	0%		A/A+	16%	82%	53%	47%	11%	0%	
AA	0%	0%	19%	23%	0%	0%		AA	0%	0%	20%	31%	2%	0%	
AAA	0%	0%	7%	0%	0%	0%		AAA	0%	0%	0%	0%	12%	0%	
MLB	0%	0%	0%	0%	0%	0%		MLB	0%	0%	0%	0%	0%	0%	
OUT	NA	7%	18%	68%	100%	100%		OUT	NA	18%	27%	22%	75%	100%	
Cum. $	*4.2k*	*13.7k*	*26.9k*	*54.6k*	*91.1k*	*128k*	*-104k*	*Cum. $*	*4.35k*	*17.1k*	*32.7k*	*47k*	*76.8k*	*113k*	*-119k*
Rookie/A-[300]	100%	57%	39%	0%	0%	0%		Rookie/A-[300]	100%	33%	34%	0%	0%	0%	
A/A+	0%	39%	61%	65%	65%	32%		A/A+	0%	67%	66%	69%	52%	43%	
AA	0%	0%	0%	4%	18%	29%		AA	0%	0%	0%	17%	15%	29%	
AAA	0%	0%	0%	2%	0%	8%		AAA	0%	0%	0%	14%	0%	0%	
MLB	0%	0%	0%	0%	0%	0%		MLB	0%	0%	0%	0%	0%	0%	
OUT	NA	4%	0%	29%	17%	30%		OUT	NA	0%	0%	0%	33%	28%	
Cum. $	*3.75k*	*9.7k*	*15.8k*	*28k*	*38.4k*	*51.3k*	*-97k*	*Cum. $*	*3.75k*	*10k*	*16.2k*	*24.5k*	*37.4k*	*49.8k*	*-98k*

Notes: Superscript numbers indicate draft position of analyzed player; shaded rows represent high school draftees; *Cum. $* = cumulative income.

performs at levels consistent with his draft position and classification. High school fielders taken at No. 10 overall also make it to the majors, though this does not occur until their fifth or sixth seasons, which is around the same age (between 22 and 23) as the 10th-overall college graduate making it in his second or third season. Both of these athletes are expected to have incomes that are roughly $1.9 million (college graduate) and $673,000 (high school) higher than their peers working outside of professional baseball. Conversely, fielders with identical biographical characteristics selected in the 300th slot are expected to have incomes far below these levels. The college graduate fielder earned an estimated $119,000 less than his counterparts in the standard workforce, while the expected income of the high school fielder was approximately $98,000 below the threshold. In a pattern similar to what was seen with the pitchers, the 300th-pick college fielder was out by year five while the high school draftee was holding out for the majors in single-A and AA leagues.

Ultimately, the analyses across both positions suggest that players wanting to break the mold of the 300th pick will have to perform at levels greater than their ideal-type averages. Looking at differences in average performance statistics between the ideal-type pitchers at No. 10 and No. 300, the latter allowed 0.15 more walks per inning, struck out 0.33 fewer batters per nine innings, and allowed 0.81 more earned runs per nine innings. The late round pitchers also pitched in far fewer innings, in lower divisions, while starting a lower percentage of their games. No. 300 fielders averaged far fewer at bats and appeared in lower classes more consistently when compared to No. 10 fielders. The later picks also averaged 0.04 more strikeouts per at bat and posted slugging averages and on-base percentages that were 7% and 3% lower, respectively.

Discussion

The aim of this article was to analyze the decision to sign a professional baseball contract by predicting the expected income of MLB draftees' over their first six seasons and comparing these figures to the incomes of non-baseball laborers. We were able to predict the career trajectories of minor league baseball players with adequate accuracy using C5.0 models. By including salary information, we were able to estimate the net incomes of baseball players over their first six seasons.

Consistent with Becker's (1975) human capital theory, many players make income-maximizing decisions by signing minor league baseball contracts. Fielders with college degrees drafted in the first 88 picks and pitchers with college degrees drafted in the first 32 picks can expect to earn between $1.1 million to $1.8 million, and $242,000 to $1 million, respectively, during the first six seasons of their careers, which was substantially more than a typical college graduate who made $231,973. Players can increase their earning potential by becoming top performers in key statistical categories, such as getting on base at higher rates or throwing more strikeouts. The average, high performing player generally remained above the income thresholds of non-baseball laborers over a longer range of draft picks and accumulated more income compared to average and low performers. College graduate fielders with a high-ranking OBP, for instance, remained above the standard income threshold for an increased range of draft picks (1–200), earning them an average of $807,137 more than a person with a four-year degree working outside of baseball. Average, high-performing pitchers were also expected to extend their income potentials over a longer range of draft picks, earning approximately $93,000 (some college) to $66,000 (college graduate) more than pitchers throwing strikeouts at a lower rate. These findings confirm those of Spurr and Barber (1994), who noted that career progression could be enhanced if one's performances exceed the mean. Our findings also support their notion that age is negatively related to the amount of time it takes to determine a player's progression from one class to another. Older draftees tend to make it to the majors, or fade out of the system, quicker than the younger players who must develop their physical and technical tools in lower levels for a longer period of time.

Considering career income alongside signing bonuses, which in 2017 ranged from $7.77 million for the first overall pick to approximately $130,000 for players selected in the tenth round (Cooper, 2017), top-300 picks—on average—are clearly making income-maximizing decisions by deciding to play professional baseball. However, many athletes are also making income-decreasing decisions when they sign. Given that picks taken after round 10 can earn bonuses of less than $10,000, many athletes are making income-decreasing decisions by playing minor league baseball. We showed that the ideal-type player selected in this area of the draft is unlikely to make it to the majors, risking an average of $97,000 to $104,000 (pitchers) or $98,000 to $119,000 (fielders) while playing in the minors. This aligns with the results of Winfree and Molitor (2007), who also showed that earnings are expected to decrease at later draft picks.

Athletes who make income-decreasing decisions to play minor league baseball seem to contradict the investment decision laid out in Becker's (1975) human capital theory. This is particularly apparent for the "middle of the pack" holdouts. Whereas earlier picks have some chance of making it to the majors, and later picks are quickly released to try other occupations, holdouts remain in lower classification levels with slim chances of moving up and negligible chances of making it to the majors; yet, they remain in the minors despite having better opportunities for income in other occupations. This was particularly noticeable among high school draftees in the scenarios we examined. Whether a product of their younger, livelier arms, or the hopeful anticipation that surrounds them, these athletes often stuck it out in MiLB to the detriment of their incomes.

An explanation for holdouts and other athletes who make wealth-decreasing decisions to play professional baseball is that baseball players are unrealistically optimistic about their careers. Psychologists suggest that people consistently overestimate the chances of good things happening to them and consistently underestimate the chances of bad things happening to them (Weinstein, 1980). Unrealistic optimism has also been studied in business and finance because it may lead to a consistent misunderstanding of risk (Malmendier & Tate, 2005). For example, entrepreneurs rate their chances of success well above the average business survival rate, with 81% seeing odds of 7 out of 10 or better and a remarkable 33% seeing odds of success of 10 out of 10 (Cooper et al., 1988). Thus, baseball players may be unrealistically optimistic about their chances of success. For example, the majority of fielders may expect to perform in the highest quartile of OBP, in which case their unrealistic optimism causes them to see professional baseball as an income-maximizing opportunity. A limitation of this explanation is other research has shown highly skilled people are much more accurate at estimating their success and can even be prone to underestimating success (Dunning et al., 2003). Therefore, future research is needed to evaluate how optimism affects athletes' career decisions.

Irrespective of athletes' optimism, it is important to note that the current minor league system depends on a labor force of excellent players who are never going to make it to the majors. The minor league ladder is based on matching prospects against equal, if not better, competition at each rung. Our findings align with previous work by Chandler and Stevens (2012) showing that at bats are one of the most important predictors of advancing through the minors. This suggests that certain players are chosen as prospects while others are chosen to be competition with less chance of advancement. Thus, we can add to the question first asked by Krautmann, et al. (2000), who pays for minor league training costs? According to Krautmann et al. (2000), owners recoup some of the minor league training costs by extracting surpluses from young MLB players who are ineligible for salary arbitration. Our analysis suggests holdouts also pay for the minor league training costs of eventual stars by serving as competition and forgoing income in another occupation. MLB should consider the situation of these workers given their role in baseball's human capital ecosystem (McLeod & Nite, 2019), especially in light of recent lawsuits and growing dissatisfaction from minor league players.

Another important point to consider is that pitchers and fielders seem to face substantially different investment decisions when deciding to play minor league baseball. This is an important finding considering most previous research has studied pitchers and fielders independently (Chandler & Stevens, 2012; Longley & Wong, 2011; Spurr & Barber, 1994; Winfree & Molitor, 2007; Witnauer et al., 2007). Regardless of whether they were drafted following college or high school, we found average fielders to have better initial income prospects than average pitchers. Because substantially more of them make it to the majors during the first six years of their careers (25.8% versus 13.2%), fielders are able to out-earn non-baseball workers over a longer range of draft picks. This aligns with prior studies suggesting that drafted fielders are more steady than pitchers (Burger & Walters, 2009), and provides another angle to the findings of Bierig et al. (2017), who noted teams should give long-term contracts to batters, rather than pitchers, if deciding between two players. Does this mean pitchers make poorer investment decisions than fielders? Future researchers should further compare pitchers and fielders to better understand career differences.

The results also make it apparent that high school and college prospects face different decisions when embarking on their careers. All else equal, high school draftees at both positions avoid sacrificing potential income over a longer range of picks. High school fielders, for instance, are able to stay above the threshold until pick 119, whereas college graduate fielders begin sacrificing income at pick 89. Similarly, high school pitchers avoid significant opportunity costs through pick 79, while college graduate pitchers forgo income as early as pick 33. Interestingly, fielders and pitchers with just some college experience appear to have the highest likelihoods of avoiding significant losses, out-earning comparable, non-baseball workers through picks 205 and 172, respectively. Taken together, these results highlight the risks that college graduates assume when they pursue a professional baseball career. High school draftees and college draftees without a four-year degree are expected to make less in the normal workforce, so they have less to sacrifice by taking a chance on baseball and can confidently enter the system at later draft picks. The more obvious difference among pitchers could also allude to the possibility that college graduate pitchers' arms are more worn down, making them more susceptible to injury and other declines in performance. Furthermore, if they initially had the talent to make the majors, they would have likely been drafted in a better position at a younger age.

Practical Implications

Our results have a series of practical implications for professional baseball players and teams. First, our models can serve as financial decision-making tools for a variety of stakeholders across the industry. For example, our estimates can be used to determine how much signing bonus or off-season work is required for players to earn the same amount they would in the normal workforce. On average, players who graduate college and forgo income by signing a contract lose approximately \$108,000 (pitchers) to \$136,000 (fielders) over their first six seasons. They could make this up with signing bonuses equal to these amounts, or an off-season job earning \$18,000 to \$23,000 a year. Pitchers who do not graduate with a four-year degree and lose money lose an average of \$43,000 (some college) to \$98,000 (high school diploma) over their first six seasons. For fielders who do not have a four-year degree, the average amounts lost are roughly \$39,468 (high school diploma) to \$55,000 (some college). College graduates have higher opportunity costs than their counterparts because they have much better prospects in the job market.

Therefore, MLB teams may have to provide larger signing bonuses to college graduates to entice them to forgo other opportunities. Adding signing bonuses to our estimates means baseball players can use our models to decide whether or not to sign a professional baseball contract. Fielders drafted in the first 89 (college graduates) or 90 picks (some college) and pitchers drafted in the first 32 picks (college graduates) or 171 picks (some college) should sign contracts regardless of signing bonus amounts. Pitchers drafted out of high school should sign contracts if they are picked in the first 78 overall picks, as should the top 118 fielders. After these cutoff points, however, the decision depends on signing bonuses. If the signing bonus is large enough to offset the sacrificed income, the player should sign. High-performing players can minimize their needs for bonuses, or generate additional income, by performing well enough to secure larger wages, even across later draft slots.

A further implication of our analysis is that fewer draftees should sign professional contracts. However, if fewer players sign contracts, fewer players would enter the labor pool, meaning less competition for MLB teams and less labor for minor league teams. Therefore, the MLB may consider changing how it pays players in order to make minor league baseball a more attractive prospect for athletes. The Blue Jays have moved in this direction, announcing a 40% to 56% pay increase across all ranks of their development system. According to The Athletic reporter, John Lott (2019), AAA players will earn $15,250 per season, AA players will earn $12,750 per season, and single-A players will earn $12,000. Inputting those salaries into our fielding models led to average six-year increases of $14,278 for fielders with some college and $5,489 for fielders with four-year degrees. The fielders with the greatest financial opportunity costs would increase their income by $18,047 (some college) and $17,599 (college graduates), or roughly $3,000 per season. However, they were originally foregoing $69,355 and $176,387, respectively, so these raises are hardly commensurate with the income they are sacrificing by staying in baseball.

Another possible solution, albeit one that creates fewer overall opportunities for aspiring players, is for MLB teams to reduce the number of minor league teams in their farm systems. According to Travis Sawchik of FiveThirtyEight Sports (2019), there are 245 MiLB affiliates spread across the 30 MLB clubs (an average of 8.2 per franchise). If a club is financially unable or unwilling to fulfill the signing bonus or wage demands of minor leaguers across eight rosters, then a reduction in teams could increase efficiency and free up the funds for the players who deserve them. This is something the Houston Astros have done in recent times, moving from nine affiliates to seven and directing their additional resources to technology, coaching, and player development. Although there may now be fewer minor leaguers in their system, the ones who were retained received better coaching (could possibly move into the upper echelon of OBP or SO9 and progress further) and greater opportunity for advancement (less competition and more visibility in the system). At bats and innings pitched, after all, are important determinants in classifying players. As our results would further suggest, even the players who get laid off when a club shrinks its farm system will likely benefit financially because they are now free to pursue non-baseball wages that would allow them to make more money than they would as holdouts in the lower ranks of professional baseball.

Limitations and Recommendations for Future Research

Predicting the simultaneous probabilities of a player reaching a given class in a given season, while still allowing for the possibility of him exiting the system altogether, is difficult. However, modern machine learning techniques, such as the C5.0 method implemented in this study, allow researchers to simultaneously predict the constrained probabilities of a player landing in any one of these categories. This is not to say that such methods are without drawbacks, though. For starters, classification models can be difficult to interpret given their size. The final trees and rulesets obtained from the models used in this study were too large to print, meaning the data had to be manipulated and simulated in order to see the effects of different predictor variables on the outcome.

It is also important to remember that C5.0 models are not linear models. They use various cutoff points and splits of different variable attributes to classify observations that match a certain pattern. Therefore, manipulating single variables, even while holding others constant, is not a straightforward task. For this reason, two stats that are known to better isolate player talent on the mound (SO9) and at the plate (OBP) were chosen. However, SO9 and OBP are just two of many possible performance stats that could have been manipulated, and their effects on their respective positions are not necessarily equivalent. Variable coefficients and marginal effects are not readily produced by C5.0 models, meaning variable importance must be judged through other, more abstract methods (e.g., the percentage of splits associated with a variable).

Using the obtained probabilities to predict player income also necessitated certain methodological liberties. For example, only the highest probable classes from the previous seasons were used as the *PriorClass* variables. The most accurate financial projection for a given season would be weighted by all of the prior classifications that had a probability of occurring within each prior season. However, the potential combinations and outcomes were too numerous to report in a single manuscript. Additionally, there were some observations in which players appeared in multiple classes throughout a season. This means that a response variable showing the highest classification they were predicted to reach might actually overstate projected income since they would not have earned the highest rate for every month that season. It also means that the performance statistics from a given season, which were weighted by at bats and innings pitched, may not have truly been representative of their performances at the higher class. For instance, a player may have accrued an above average on-base percentage and been classified as having reached AAA, whereas the majority of his at bats actually occurred in AA. Future attempts to catalogue the classification patterns of professional baseball players could devise more intricate ways of addressing this limitation.

Continuing, future studies could also conduct a more thorough evaluation of the *HalfSeason* (Rookie and A-) and *FullA* (A and A+) aggregations that were used as levels in the *Class* outcome variable. We combined these classes because of their similar playing schedules and pay scales, and to lend convenience and accuracy to the classification process; nonetheless, the intricacies that differentiate these divisions were ignored, meaning the prospects of players appearing in certain leagues may be overestimated or underestimated. For example, rookie leagues such as the

Dominican Summer League, the Arizona League, and the Gulf Coast League are considered inferior to advanced rookie leagues such as the Appalachian League and the Pioneer League, and have different weather conditions and fewer spectators. However, these athletes also share in common that they only play a half-season and are paid according to the same wage scale. This raises the point that the C5.0 models could also include additional predictor variables beyond those selected for this analysis. We studied prior literature and included those that we felt were necessary for making the initial models, but there are additional variables, such as those related to the organization drafting the player (e.g., team payroll and past and current performance), the minor league team he is assigned to (e.g., weather and reputation), college stats, and high school stats, that could be collected and included in future analyses. These variables could lead to more accurate projections and help the models find additional utility among prospective players, agents, coaches, and general managers.

Conclusion

We used C5.0 machine learning techniques to model the flow of MLB draftees through minor and major league classifications over the first six years of their careers. We then combined the results with financial data to calculate the expected income of fielders and pitchers, and assessed their financial opportunity costs in relation to workers who pursued careers outside of baseball. The results showed that, all else equal, fielders drafted in the first 89–90 picks and pitchers drafted in the first 32 picks (college graduates) or 171 picks (some college) should sign contracts. Pitchers drafted out of high school should sign contracts if they are picked in the first 78 picks, as should fielders selected in the top 118 picks. After these cutoff points, however, signing bonuses are needed to make up for lost income, and teams are currently exploring ways to provide these additional funds. High-performing players, as evidenced by ranking above the 75th percentile for SO9 (pitchers) and OBP (fielders), are able to limit their opportunity costs over a longer range of draft picks and accumulate higher income than average and low performing players. Moving forward, researchers should incorporate more variables and enhance the C5.0 models for use in answering additional theoretical and practical questions about player progression through professional baseball.

Declaration of Conflicting Interests

The author(s) declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Funding

The author(s) received no financial support for the research, authorship, and/or publication of this article.

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Notes

1. After six seasons in the minors, players not on a 40-man MLB roster become free agents.
2. Performance statistics of high school/non-NCAA Division I players are not readily available.
3. Given the primary focus of our study, we valued a method that efficiently and accurately classified a large sample of MLB draftees—each of whom can appear in several different classifications via patterns of movement that are not always sequential—over methods that allow for easier interpretation of the predictor variables. Prior studies from which we drew several of the variables used in our C5.0 models have looked more exclusively at the effects certain variables have on direct promotion from one class to another (e.g., Bradbury, 2010; Chandler & Stevens, 2012; Longley & Wong, 2011; Spurr & Barber, 1994).
4. To be eligible for the MLB First-Year Player Draft, a player must have graduated high school, finished his junior or senior year at a four-year college, or be enrolled in junior college. However, the *College* variable only indicated whether a player attended college for any length of time and did not differentiate between these different scenarios. As such, the non-baseball income had to include the expected income for high school graduates, four-year college graduates, and those who had taken some college. The calculations for 18–24 year old males in the first two categories were available from the U.S. Census Bureau (2017). The estimate for an 18–24 year old male with "some college" was also available, but players with "some college" could have obtained an associate's (two-year) degree in junior college or by their junior year at a four-year institution. Therefore, the estimate for "some college" individuals in this study was created by averaging the U.S. Census Bureau's (2017) income for 18–24 year old males with "some college" with those who had associate's degrees, weighted by the estimated number of 18–24 year old males present in the two categories.

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